

Simone Carletti, M.Sc.

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🌐 <https://simonecarletti.github.io/>



About Me

Computer Engineering M.Sc. student specializing in robotics and autonomous systems. Hands-on experience in control systems, perception algorithms, ROS2, embedded software, and full-stack web development. Skilled in C++/Python/Typescript, with research experience at DTU and EPFL. Eager to contribute as a Robotics Engineer in innovative R&D projects.

Employment History

- 2020 – now **Full-Stack Developer**, Genomedics S.r.l
Angular (Typescript, SCSS, HTML), PHP, Laravel, SQL, Python.
Lead developer of three key projects that handle the release, deployment and licence management for all the company's products. Developer for the company's main project (K-Platform).
- 2023 – 2024 **Control Engineer**, DIANA student team.
Development of a martian rover for the ERC competition
- 2018 – 2022 **Web developer**, Scuolafacile, ITIS Meucci.
Development of a platform that allows schools to manage timetables and substitute teachers

Education

- 2025 – 2026 **Master's Thesis**, École Polytechnique Fédérale de Lausanne (EPFL).
L1-Adaptive augmentation for robust morphing-wing drone flight control.
Skills: Control Systems, C++, Python, Simulink, ROS2, Git, Linux
- 2024 – 2025 **M.Sc. Autonomous Systems**, Technical University of Denmark.
1 year Erasmus+ exchange.
Courses (and relevant projects):
- **Building Dependable Robot Systems:** design of algorithms and control systems with the *Robot* platform for the DTU RoboCup competition
 - **Digital Control:** design of digital controllers and implementation on embedded microcontrollers for a differential drive robot
 - **Perception for Autonomous Systems:** computer vision, 3D cloud processing, visual odometry
 - **Deep Learning:** FFNN, CNN, Autoencoders, Transformers, GAN, PyTorch.
 - **Advanced topics in Reinforcement Learning**
 - **Image Analysis:** introduction course with application in the medical field

Education (continued)

2023 – now

■ **M.Sc. Computer Engineering**, Politecnico di Torino.
Automation and Intelligent Cyber-Physical Systems - CURRENT GPA: 29.48/30

Courses (and relevant projects):

- **Sensors, embedded systems and algorithms for Service Robotics**: course on mobile robotics. Implementation of an EKF for localization in ROS2, using Python
- **Robotics**: basics of robotic manipulators. Development on Matlab.
- **Nonlinear control and aerospace applications**: implementation of a series of control architectures with application in spacecraft maneuvers
- **Estimation, Filtering and System Identification**
- **System and Device Programming**: implementation of kernel functionalities (OS161); Development of a desktop program with Rust
- **Computer architectures**: development of Quoridor game (link) on a LandTiger board with multiplayer capabilities over CAN
- **Modeling and control of cyberphysical systems**: distributed control of a multi-agent system of 7 magnetic levitators
- **Data Science and Database Technology**
- **Big data processing and analytics**

2020 – 2023

■ **B.Sc. Computer Engineering**, Università degli Studi di Firenze.
GRADUATED WITH 110/110 CUM LAUDE (WITH HONORS)

Thesis title: *Creation of a sensor network to monitor the greenhouses of the Botanical Garden of Florence*

Free choice courses: Discrete mathematics and code theory, Financial mathematics

2019

■ **CyberChallenge**, Cybersecurity course at University of Florence.
The first italian cybersecurity training program for talented university and high school students

Volunteering

2025 – now

■ **ESN Copenhagen & ESN DTU**: Organization of events, support for international students, and promotion of cultural exchange.

Skills

Languages ■ Strong reading, writing and speaking competencies for English. Native italian speaker.

Coding ■ Typescript, Javascript, C, C++, Python, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly (x86, MIPS), LaTeX

Frameworks ■ Angular, Vue.js, Laravel, Simulink, ROS2

Misc. ■ Git, Linux (full-time user with experience in various fields), Raspberry-Pi, Arduino, Photography (digital and analog cameras), Guitar, Soldering, ESP8266, NodeMCU, PID Control, SLAM, Sensor Fusion, Trajectory Planning, Localization

Certifications and Achievements

- 2023  **Cambridge B2 (FCE).** *ID: C4570372*
- 2018  **ECDL Full Standard.** *ID: IT 2156394*
- 2020 – 2023  **#TuttoMeritoMio Excellence Scholarship** Winner of #TuttoMeritoMio, a scholarship dedicated to the top 100 students of the University of Florence